YEAR 11 PHYSICS, AQA GCSE, LESSON 9 OF 10

Atoms & Nuclear Radiation.

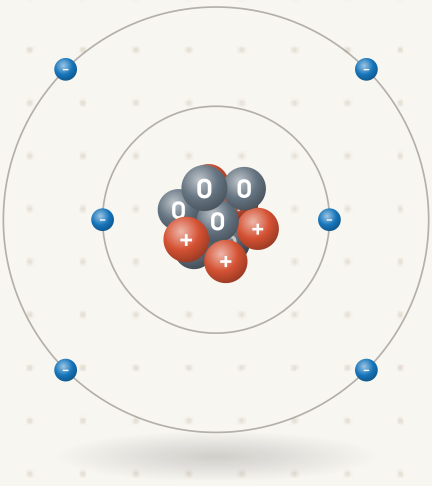
Lesson 09 Half-life calculations

A hospital uses iodine-131, with a half-life of 8 days. How long before less than 1% of it is left, and why does that matter?

IN THIS LESSON YOU WILL

- count half-lives to find how much is left
- find a half-life from data
- find how long it takes to fall to a given value
- work out the net decline as a ratio

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lesson 9** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this lesson is highlighted)

01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations
Counting half-lives, finding a half-life from data, net decline as a ratio (slides 36)

10 Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

09 Half-life calculations.

AQA Physics 4.4.2.3; Combined Science 6.4.2.3

BIG QUESTION

A hospital uses iodine-131, with a half-life of 8 days. How long before less than 1% of it is left, and why does that matter?

BY THE END YOU CAN

- count half-lives to find how much is left
- find a half-life from data
- find how long it takes to fall to a given value
- work out the net decline as a ratio

half-life | number of half-lives | initial activity | fraction remaining | net decline | ratio

SLIDES AND SIMULATIONS

Teaching deck, slide 36

Using half-life: from a graph, counting halvings, and as a ratio.

panphy.app/gcsephy/decks/atoms-and-radiation/#36

RETRIEVE

Do now

From earlier lessons. No notes!

1 Define half-life.

The time for the number of unstable nuclei (or the count rate) to halve.

2 A count rate falls from 400 to 200 in 8 days. What is the half-life?

8 days.

3 Why is radioactive decay described as random?

You cannot predict which nucleus will decay, or when.

EXPLAIN

Counting the halvings

Three steps

1. Work out the **number of half-lives**: time ÷ half-life. Use the same units for both.
2. **Halve** the starting value that many times.
3. Give the answer **with a unit**.

WORKED EXAMPLE

A source has an activity of 640 Bq. Its half-life is 2 hours. Calculate its activity after 8 hours.

HALF-LIVES	8 ÷ 2 = 4
HALVE	640 → 320 → 160 → 80 → 40
ANSWER	**40 Bq**

YOUR TURN

A source has an activity of 3200 Bq. Its half-life is 5 days. Calculate its activity after 15 days.

HALF-LIVES	15 ÷ 5 = 3
HALVE	3200 → 1600 → 800 → 400
ANSWER	400 Bq

Finding the half-life from data

WORKED EXAMPLE

The activity of a sample falls from 2400 Bq to 300 Bq in 36 minutes. Calculate the half-life.

HALVINGS	2400 → 1200 → 600 → 300: 3 halvings
HALF-LIFE	36 ÷ 3
ANSWER	**12 minutes**

YOUR TURN

A count rate falls from 960 to 60 counts per minute in 20 hours. Calculate the half-life.

HALVINGS	960 → 480 → 240 → 120 → 60: 4
HALF-LIFE	20 ÷ 4
ANSWER	5 hours

Net decline as a ratio

After n half-lives, the fraction left is $(½)^n$. The **net decline** is how much the activity has fallen. As a ratio of the initial activity, it is $1 - (½)^n$.

Example: 200 Bq falls to 50 Bq in 2 half-lives. Net decline = 200 − 50 = 150 Bq. As a ratio of the initial activity: 150 : 200 = **3 : 4**, or ¾.

Half-lives	Fraction left	Net decline
1	½	½
2	¼	¾
3	⅛	⅞
4	1/16	15/16

PRACTISE

Try it

1 SUPPORTED A source has an activity of 1200 Bq and a half-life of 6 hours. Calculate its activity after 24 hours.

HALF-LIVES	24 ÷ 6 = 4
HALVE	1200 → 600 → 300 → 150 → 75
ANSWER	75 Bq unit:

2 ON YOUR OWN How long does it take for the activity of a source to fall from 1000 Bq to 125 Bq, if its half-life is 6 hours?

1000 → 500 → 250 → 125 is 3 half-lives; 3 × 6 = 18 hours.

3 Use the data to find the half-life. (Hint: find when the count rate is half of 800.)

Time / min	0	10	20	30	40
Count rate / counts per minute	800	566	400	283	200

800 halves to 400 at 20 minutes (and 400 → 200 from 20 to 40 min), so the half-life is 20 minutes.

4 Explain why Mia is wrong. What fraction is really left after 20 minutes?

Mia: The half-life is 10 minutes, so the activity will be zero after 20 minutes.

Each half-life only halves the activity; it never jumps to zero. 20 minutes is 2 half-lives, so a quarter (¼) is left.

5 A source has an initial activity of 1200 Bq. Calculate its activity after 3 half-lives, and the net decline as a ratio of the initial activity.

1200 → 600 → 300 → 150 Bq. Net decline = 1200 − 150 = 1050 Bq; 1050 : 1200 = 7 : 8 (⅞).

6 CHALLENGE Iodine-131 has a half-life of 8 days. (a) What fraction is left after 32 days? (b) How many whole half-lives until less than 1% is left? How many days is that?

(a) 32 ÷ 8 = 4 half-lives: 1/16 (6.25%) is left. (b) $(½)^6$ = 1.6%, $(½)^7$ = 0.78%, so 7 half-lives = 56 days.

Get the marks: Show the halvings, not just the answer. Units: keep kBq as kBq.

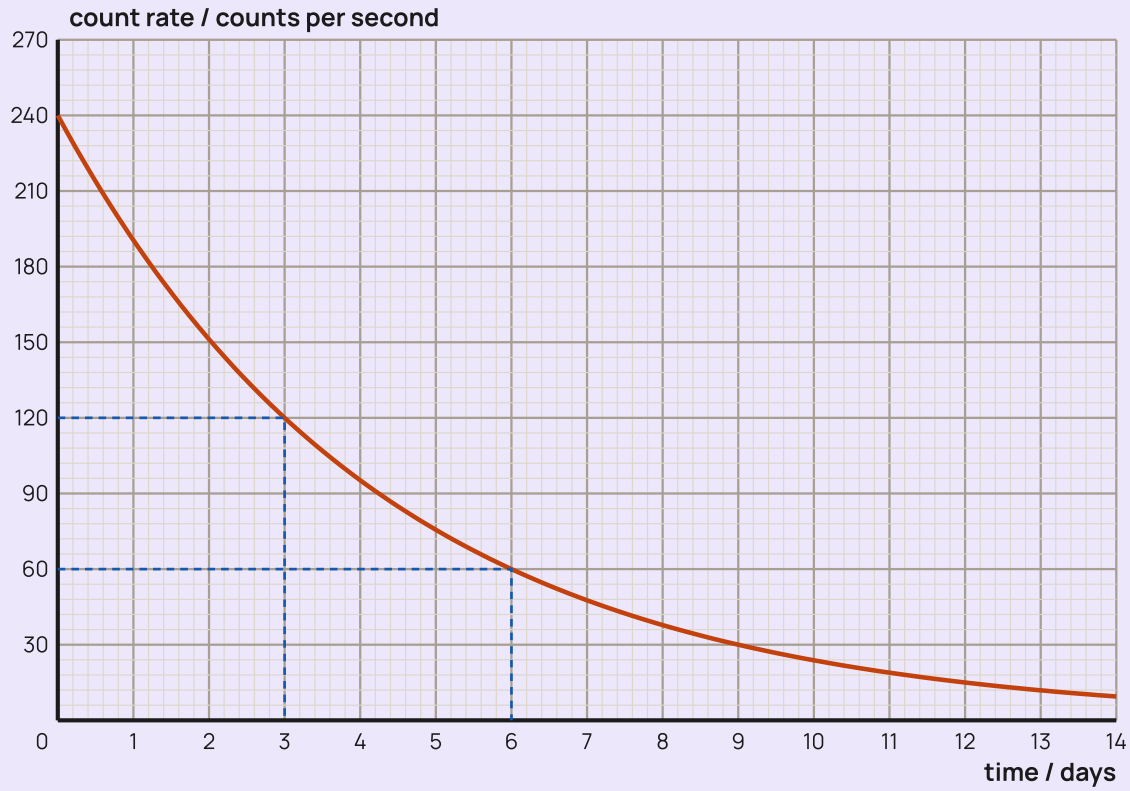
7 Cobalt-60 has a half-life of 5.3 years. A source has an activity of 400 kBq. Calculate its activity after 15.9 years. **[2 marks]**

15.9 ÷ 5.3 = 3 half-lives (1); 400 → 200 → 100 → 50 kBq (1).

8 Calculate the net decline in the activity of the cobalt-60 source after 15.9 years, as a ratio of its initial activity. **[2 marks]**

Net decline = 400 − 50 = 350 kBq (1); 350 : 400 = 7 : 8 (or ⅞) (1).

9 The graph shows the count rate from a sample. Determine the half-life. Show on the graph how you found it. **[2 marks]**



240 → 120 at 3 days, lines drawn on the graph (1); half-life = 3 days (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Number of half-lives = time ÷ half-life (same units).
- Halve the starting value once for each half-life.
- Half-life from data: count the halvings, then divide the time by that number.
- After n half-lives, $(½)^n$ is left.
- HT: net decline = initial − final; as a ratio of the initial, $1 - (½)^n$.

COVER THE PAGE AND ANSWER

1 What fraction is left after 3 half-lives?

2 800 Bq falls to 100 Bq. How many half-lives?

3 Half-life 4 h. How long for 640 Bq to fall to 80 Bq?

4 HT: net decline after 2 half-lives, as a ratio?

I CAN…

- calculate how much is left after a number of half-lives
- find a half-life from data or a graph
- find how long a source takes to fall to a value
- express the net decline as a ratio

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

Nuclear equations

Lesson 6

Emitted	Symbol	Mass no.	Atomic no.
alpha	$^{4}_{2}\mathrm{He}$	− 4	− 2
beta	$^{0}_{-1}\mathrm{e}$	no change	+ 1
gamma	γ	no change	no change
neutron	$^{1}_{0}\mathrm{n}$	− 1	no change

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Properties of radiation

Lesson 7

	Alpha	Beta	Gamma
Ionising	strongly	moderately	weakly
Range in air	a few cm	about 1 m	very far
Stopped by	paper, skin	a few mm of aluminium	reduced by thick lead or concrete
Typical use	smoke alarm	thickness gauge	medical tracer

More ionising → less penetrating.

Half-life from a graph

Lesson 8

1. Read the start value.
2. Halve it.
3. Across to the curve, down to the time axis.
4. Check with a second halving.

Half-life calculations

Lesson 9

- Number of half-lives = time ÷ half-life.
- After n half-lives, $(½)^n$ is left: ½, ¼, ⅛, 1/16…
- Net decline = initial − final; as a ratio of the initial, $1 - (½)^n$: ½, ¾, ⅞…

REFERENCE

Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Net decline: How much the activity has fallen; often given as a ratio of the initial activity.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. The time for the number of unstable nuclei (or the count rate) to halve.
2. 8 days.
3. You cannot predict which nucleus will decay, or when.

COVER THE PAGE

1. ⅛
2. 3
3. 3 half-lives = 12 hours.
4. ¾ (3 : 4)